PARTICLE-RESOLVED PERIODIC ORBITS AND SHARP RECURRENT-CORE ENTRY FOR RULE 184 ON FINITE RINGS

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Abstract. We study elementary cellular automaton Rule 184 on a binary ring of length n, retaining the conserved particle number. The classical asymptotic description says that every orbit reaches either the cyclic no-11 shift, where particles translate right, or the cyclic no-00 shift, where holes translate left. We refine this finite-ring picture in two directions. On the layer with m particles, the largest first-entry time into this recurrent core is exactly

$$\left(\min\{m, n-m\} - 1\right)_+.$$

The proof uses a min-plus formula for labeled particle positions. We also derive the particle-weighted iterate-fixed polynomial. If $d = \gcd(n, k)$, $r = n/d$, and $H(d, j) = \frac{d}{d-j}\binom{d-j}{j}$ counts j-subsets of a labeled cycle, then

$$\sum_{F_n^k x = x} u^{|x|} = \sum_{j=0}^{\lfloor d/2 \rfloor} H(d, j)(u^{rj} + u^{n-rj}) - 2\mathbf{1}_{2|d} u^{n/2}.$$

Consequently $\#\operatorname{Fix}(F_n^k) = 2L_d - 2\mathbf{1}_{2|d}$, where L_d is a Lucas number. Möbius inversion gives every temporal orbit, its particle-number refinement, and the finite-map Artin–Mazur zeta function. Exact enumeration checks all stated ledgers on the registered finite ranges.

1. Introduction and ownership boundary

Rule 184 is the number-conserving elementary cellular automaton obtained from Wolfram's binary local-rule numbering [8]. It is also the parallel, deterministic, unit-speed traffic rule: a particle moves one site to the right precisely when that site is empty. Nishinari and Takahashi connected the rule to an ultradiscrete Burgers equation and derived its traveling-wave asymptotics [6]. Belitsky and Ferrari treated invariant measures and convergence [1], while Fukś developed exact preimage methods and the density-classification use of Rule 184 [2]. Recent work studies space–time jam clusters, relaxation observables, and their height-function description in the same traffic model [3, 4].

We therefore take as owned background the traffic interpretation, particle conservation, density transition, no-11/no-00 asymptotic phases, and the fact that those phases translate. The residual finite-ring questions here are narrower: the exact worst first-entry time at each conserved particle number, and the complete temporal orbit ledger refined by that number.

The main results are:

(1) a sharp layerwise transient bound, proved by a closed min-plus solution for labeled car positions;
(2) a particle-weighted formula for every $\operatorname{Fix}(F_n^k)$;
(3) exact-period and exact-particle orbit counts, hence a finite temporal Artin–Mazur zeta product;
(4) the microcanonical exponential rate of recurrent states.

No claim is made to the discovery of Rule 184, its recurrent phases, the Lucas enumeration of the golden-mean shift, or Möbius inversion. The zeta below is temporal for a finite, noninvertible cellular map; it is not the spatial zeta of a frozen subshift.

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2. The recurrent core

Fix an integer $n \geq 1$. Let $\mathbb{Z}_n = \mathbb{Z}/n\mathbb{Z}$ and write $x = (x_i)_{i \in \mathbb{Z}_n} \in \{0,1\}^{\mathbb{Z}_n}$. We use the local rule

$$(F_n x)_i = x_i x_{i+1} + x_{i-1}(1 - x_i). \tag{1}$$

The two summands cannot both equal one. The first records a particle blocked by its right neighbor; the second records a particle entering from the left. Thus $|F_n x| = |x|$, where $|x| = \sum_i x_i$.

Let X_n^{11} be the cyclic words avoiding 11, and let X_n^{00} avoid 00. On X_n^{11} every particle moves right, so F_n is right rotation. On X_n^{00} every hole moves left, so F_n is left rotation. The standard finite-ring convergence theorem gives

$$\operatorname{Rec}(F_n) = X_n^{11} \cup X_n^{00}. \tag{2}$$

The next section reproves entry into this set while sharpening its time to an exact layerwise extremum. Since F_n restricts to rotations on the two pieces, entry also proves that no point outside the union is recurrent.

The intersection is empty for odd n. For even n it consists of the two alternating configurations; Rule 184 interchanges them.

3. A sharp first-entry time

Define

$$\tau_n(x) = \min\{t \geq 0 : F_n^t(x) \in X_n^{11} \cup X_n^{00}\}.$$

Lemma 3.1 (min-plus particle formula)**.** *Suppose x has $m \geq 1$ particles. Label their lifted positions so that*

$$p_j(0) < p_{j+1}(0), \qquad p_{j+m}(0) = p_j(0) + n.$$

Then, for every $t \geq 0$,

$$p_j(t) = \min_{0 \leq s \leq t} (p_{j+s}(0) + t - 2s). \tag{3}$$

Proof. A particle either advances one site or is stopped one site behind the next particle, hence

$$p_j(t+1) = \min\{p_j(t) + 1, p_{j+1}(t) - 1\}.$$

Substitution of the formula at time t makes the first minimum range over $0 \leq s \leq t$ and the second over $1 \leq s \leq t+1$; their union is exactly (3) at time $t+1$. ☐

Theorem 3.2 (sharp layerwise entry depth)**.** *For $0 \leq m \leq n$,*

$$\max_{|x|=m} \tau_n(x) = \left(\min\{m, n-m\} - 1\right)_+. \tag{4}$$

Proof. First assume $1 \leq m \leq n/2$. Put $a_j = p_j(0) - 2j$. Then $a_{j+m} = a_j + n - 2m \geq a_j$. At time $m-1$, define

$$b_j = \min_{0 \leq s \leq m-1} a_{j+s}.$$

The window defining b_{j+1} drops a_j and adds $a_{j+m} \geq a_j$, so $b_{j+1} \geq b_j$. By lemma 3.1,

$$p_{j+1}(m-1) - p_j(m-1) = 2 + b_{j+1} - b_j \geq 2.$$

Every particle is therefore isolated, and $F_n^{m-1}(x) \in X_n^{11}$.

For the word $1^m 0^{n-m}$, with occupied positions initially $0, \ldots, m-1$, a direct induction gives, for $0 \leq t \leq m-1$, the occupied set

$$\{0, \ldots, m-t-1\} \cup \{m-t+1, m-t+3, \ldots, m+t-1\}. \tag{5}$$

The second set is empty at $t = 0$. Indeed, one application of the traffic rule shortens the first block at its right end and advances every already detached particle by one. The assumption $n \geq 2m$ leaves a cyclic gap of at least two after the last displayed particle. Thus an adjacent pair remains for every $t < m-1$, whereas at $t = m-1$ all particles are isolated. The bound is attained. The cases $m = 0, 1$ have entry time zero.

For completeness, define

$$(\Theta x)_i = 1 - x_{-i}, \qquad i \in \mathbb{Z}_n.$$

A substitution in (1) gives $F_n\Theta = \Theta F_n$; moreover, Θ exchanges avoidance of 11 and avoidance of 00, and sends the m-particle layer to the $(n-m)$-particle layer. Hence $\tau_n(\Theta x) = \tau_n(x)$, which proves the high-density half of (4). $\square$

The global worst depth is therefore $(\lfloor n/2 \rfloor - 1)_+$. Notice that theorem 3.2 concerns first entry into the recurrent core at fixed density; it is not a distributional jam-lifetime statement.

4. Particle-weighted iterate-fixed counts

For $d \geq 1$ and $0 \leq j \leq \lfloor d/2 \rfloor$, set

$$H(d,j) = \frac{d}{d-j}\binom{d-j}{j}, \tag{6}$$

with $H(d,0) = 1$. Cutting a labeled cycle immediately after a marked vertex, or using the standard cycle/path decomposition, shows that $H(d,j)$ counts cyclic binary words of length d with j ones and no adjacent ones.

Theorem 4.1 (weighted fixed polynomial)**.** *Let $d = \gcd(n,k)$ and $r = n/d$. Then*

$$\Phi_{n,k}(u) := \sum_{F_n^k x = x} u^{|x|} = \sum_{j=0}^{\lfloor d/2 \rfloor} H(d,j)(u^{rj} + u^{n-rj}) - 2\mathbf{1}_{2|d} u^{n/2}. \tag{7}$$

If $L_0 = 2, L_1 = 1$ and $L_{s+1} = L_s + L_{s-1}$, then

$$\#\,\mathrm{Fix}(F_n^k) = 2L_{\gcd(n,k)} - 2\mathbf{1}_{2|\gcd(n,k)}. \tag{8}$$

Proof. An iterate-fixed point is recurrent, hence lies in (2). On X_n^{11}, F_n^k is rotation by k. Such a configuration is the r-fold repetition of a labeled cyclic word of length d; if the base word has j particles, it contributes $H(d,j)u^{rj}$. Particle–hole reflection gives the term $H(d,j)u^{n-rj}$ from X_n^{00}.

The two alternating configurations occur in both sums exactly when they are fixed by the kth iterate. Rule 184 swaps them, so this requires k even. When n is even, k is even exactly when $d = \gcd(n,k)$ is even; if d is even then n is automatically even. Their common weight is $n/2$, and each of the two points was counted once in each branch, giving the displayed subtraction by two. Finally, $\sum_j H(d,j) = L_d$, the cyclic golden-mean count, which yields (8). $\square$

Thus all unweighted temporal fixed data collapse to the gcd and one parity bit, while $\Phi_{n,k}$ retains the conserved-particle decomposition.

5. Exact temporal orbits and zeta

Let μ be the number-theoretic Möbius function. For $\ell, j \geq 0$, put

$$P(\ell,j) = \sum_{e|\gcd(\ell,j)} \mu(e) H\left(\frac{\ell}{e}, \frac{j}{e}\right). \tag{9}$$

This is the number of labeled no-11 cyclic words having least rotation period ℓ and j ones, by ordinary divisor Möbius inversion [7].

Theorem 5.1 (orbit ledger)**.** *Let $c_{n,\ell}$ be the number of temporal Rule-184 orbits of exact length ℓ. Then*

$$c_{n,\ell} = \begin{cases} \dfrac{2}{\ell}\displaystyle\sum_{e|\ell} \mu(\ell/e) L_e - \mathbf{1}_{\ell=2}, & \ell \mid n, \\ 0, & \ell \nmid n. \end{cases} \tag{10}$$

Consequently the temporal Artin–Mazur zeta function is

$$\zeta_{F_n}(z) = \exp\left(\sum_{k\geq 1} \frac{\#\operatorname{Fix}(F_n^k)}{k} z^k\right) = \prod_{\ell\mid n}(1-z^\ell)^{-c_{n,\ell}}. \tag{11}$$

More precisely, set $r = n/\ell$. If $m = rj < n/2$, the number of length-ℓ orbits with m particles is $P(\ell, j)/\ell$; the same formula holds for $m = n - rj > n/2$. At half filling there is exactly one orbit, of length two, when n is even. All other particle-resolved entries vanish.

Proof. Every temporal period divides n because both recurrent branches are rotations of an n-ring. For $\ell \mid n$, Möbius inversion of (8) gives the number of exact-period points. The parity term is explicit:

$$\sum_{e\mid\ell} \mu(\ell/e)\mathbf{1}_{2\mid e} = \mathbf{1}_{\ell=2}.$$

For odd ℓ the sum is empty; for $\ell = 2s$ it equals $\sum_{f\mid s}\mu(s/f)$, which is one only at $s = 1$. Hence the correction to exact-period points is $-2\mathbf{1}_{\ell=2}$, and division by ℓ gives the term $-\mathbf{1}_{\ell=2}$ in (10). The standard finite-map fixed-point-to-orbit identity then gives (11) [5].

For the refinement, a low-density configuration of least temporal period ℓ is the r-fold repetition of a primitive cyclic word counted by (9). Divide its labeled points into rotation orbits of size ℓ. Reflection and complementation give the high-density statement. The two alternating descriptions coincide at half filling, producing one two-cycle rather than two. □

6. Recurrent-state entropy by density

At particle number m, (2) and the intersection correction give

$$R_{n,m} = H\big(n, \min\{m, n-m\}\big).$$

Along any sequence of integer pairs with $n \to \infty$ and $m/n \to \rho$, let $\alpha = \min\{\rho, 1-\rho\} \in [0, 1/2]$, Stirling's formula gives

$$\lim_{\substack{n\to\infty\\ m/n\to\rho}} \frac{1}{n}\log R_{n,m} = (1-\alpha)\log(1-\alpha) - \alpha\log\alpha - (1-2\alpha)\log(1-2\alpha). \tag{12}$$

The endpoint convention is continuous. Differentiation shows that the two density-symmetric maxima occur when $\alpha = (5-\sqrt{5})/10$, and their value is $\log\varphi$, with $\varphi = (1+\sqrt{5})/2$. This is a microcanonical count of recurrent states, not a new version of the traffic flux diagram.

7. Exact control and scope

The accompanying standard-library Python program independently implements the local rule, functional graphs, cyclic hard-core counts, min-plus particle motion, Möbius inversion, and the closed formulas. It exhausts recurrent cores, the particle–hole reflection conjugacy, and sharp layer depths for $n \leq 14$; checks the all-time min-plus formula through $t = 2n$ for every low-density state with $n \leq 12$; verifies all weighted fixed polynomials for $n \leq 12$ and $k \leq 2n$; and checks particle-resolved temporal cycles for $n \leq 13$. The control guards conventions and endpoints; the proofs above carry the all-size claims.

The result is a finite-ring exact closure. Its first-entry variable τ_n is the first time the full configuration reaches the translating recurrent core. It is not the lifetime of a chosen jam cluster, nor an ensemble relaxation statistic of the kind studied in [3, 4]. The paper does not address noisy traffic, infinite-volume mixing rates, hydrodynamic limits, or priority beyond the bounded owner audit described in the introduction. External release remains on hold.

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